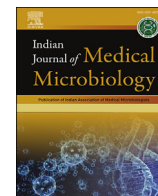




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## Case Report

Report of a carbapenemase gene *bla*<sub>IMP-4</sub> in multi-drug resistant *Escherichia coli* from sewage water: A threat on clinical-environmental interphaseDeepjyoti Paul<sup>1</sup>, Nurjhan Begum Mazumder<sup>1</sup>, Jayalaxmi Wangkheimayum, Amitabha Bhattacharjee<sup>\*</sup>

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## ABSTRACT

Acquired carbapenemases pose a significant role in the dissemination of antimicrobial-resistant Enterobacteriaceae and in this study we have identified the occurrence of *bla*<sub>IMP-4</sub> in *E. coli* isolate from a sewage outfall located nearby a secondary health Centre. It was found to co-existed with *bla*<sub>CTX-M-15</sub> located within a self-conjugable plasmid of IncF type. The current study underscores environment as a potential reservoir of carbapenem resistance and the need of the hour is to track and check dissemination of resistance in environment, human and agricultural settings.

## 1. Introduction

Imipenem resistant carbapenemase (IMP) encodes for *bla*<sub>IMP</sub> gene is a major plasmid-mediated carbapenemase among multi-drug resistant Enterobacteriaceae and is one of the most geographically widespread metallo-beta-lactamase in worldwide. IMP is also one of the most recorded carbapenemases prevalently appeared in community acquired and health care associated infections and different variants of this resistance determinant is being reported [1]. The analysis of sewage sample is a good proxy to understand the prevalence of a resistance mechanism in the community as well as in the environment as waste water sample is considered as the significant carrier of industrial wastage, human wastes from different home activities and carrying multiple pathogens such as bacteria, parasitic worms as well as organic and inorganic particles. *Escherichia coli*, a ubiquitous organism, has been listed as a critical concern by the World Health Organisation because it is universally carried in the human gut and its commonest cause of Urinary tract infections (UTI's) in the community and in hospitals.

## 2. Case report

The present study describes the detection of *bla*<sub>IMP-4</sub> in an *E. coli* isolate obtained from a sewage outfall located nearby a secondary health Centre of Southern Assam, India. This study was carried out for a duration of 3 months lasted from November 2019 to February 2020 and the

samples were collected randomly from different sewage sites located in the nearby areas (approximately 500 m) of different health Care Centre's of southern Assam, India and a total of 104 *E. coli* isolates were obtained and characterized for further study. *E. coli* isolates were selected in brilliant green lactose-bile broth (BGLB) medium containing cefotaxime (30mg/ml) and vancomycin (50mg/ml) to screen out the resistant organism and also to inhibit the growth of any Gram-positive bacteria. The isolates were identified and susceptibility testing was performed using VITEK-2 compact automated system (bioMérieux, Marcy-l'Étoile, France). The co-existence of ESBL (*bla*<sub>CTX-M-15</sub>, *bla*<sub>SHV</sub>, *bla*<sub>OXA-2</sub>, *bla*<sub>OXA-10</sub>, *bla*<sub>TEM</sub>) [2], carbapenemases (*bla*<sub>NDM</sub>, *bla*<sub>VIM</sub>, *bla*<sub>IMP</sub>, *bla*<sub>KPC</sub>, *bla*<sub>OXA-48</sub>) [2], aminoglycoside (*armA*, *rmtA-rmtD*, *npmA*, *aadB*, *aadA1*, *aac3*) [3] and quinolone (*qnrA*, *qnrB*, *qnrS*, *qnrD*, *qnrC* and *aac(6')-Ib-cr*) [4] resistance determinants were further genotypically characterized using PCR assay.

ESBL resistant determinants were amplified using the following reaction condition, initial denaturation at 95 °C for 5 mins, 32 cycles at 95 °C for 1min, 54 °C for 1min, 72 °C for 1mins and final extension at 72 °C for 7 mins and carbapenemase resistant genes were targeted using initial denaturation at 94 °C for 5 mins, 32 cycles at 94 °C for 30s, 53 °C for 1min, 72 °C for 1mins and final extension at 72 °C for 5 mins. For amplifying the quinolone resistant determinants, the reactions were run at the condition of initial denaturation at 95 °C for 2 min, 35 cycles of 95 °C for 50 s, 95 °C for 40 s and 95 °C for 1.20 min, and a final extension at 95 °C for 5 min whereas the reactions were performed under the following

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conditions; initial denaturation at 94 °C for 5 min, 34 cycles of 94 °C for 30 s, 52 °C for 40 s, and 72 °C for 1 min 20 s, and a final extension at 72 °C for 7 min for amplifying aminoglycoside resistant determinants.

PCR amplified products were further sequenced to confirm the presence of resistant gene and the variants and analysed using the blast tool of the National Center for Biotechnology Information. PCR and sequencing confirmed that an isolate was harboring *bla*<sub>IMP-4</sub> and was also carrying *bla*<sub>CTX-M-15</sub>, *rmtA*, *rmtD*, *aadB*, *aadA1*, *aac3*, *aac(6')*-Ib, *aph(2')*-Ic and *aph(3//)-I* and *sul1*. Transfer of this *bla*<sub>IMP-4</sub> resistance gene was checked by transformation (heat shock method) and conjugation (*E. coli* J53 as recipient) assays and both the assays confirmed the horizontal transfer of *bla*<sub>IMP</sub> when selected in a medium containing 0.5mg/L of imipenem. Furthermore, the PCR assay of the transconjugants confirmed the presence of *bla*<sub>CTX-M-15</sub> gene within the same plasmid of approximately 90–100kb in size. Incompatibility typing of the conjugable plasmid encoding both *bla*<sub>IMP</sub> and *bla*<sub>CTX-M-15</sub> was determined using PCR based replicon typing targeting 22 different replicon types [5] and the PCR results exhibited that both the resistance determinants were carried within a single transconjugable IncFIC type plasmid. The sequence typing of the *E. coli* strain was determined by multi-locus sequence typing but the isolate did not match with any of the existing ST-types in the database and was assigned as unknown sequence type.

### 3. Discussion

Carbapenemase producing bacteria are now spreading all over the world and the emergence of these resistance determinants within Gram negative rods with the continuing evolving of different carbapenemases types has threaten the public health system worldwide. IMP, one such diverse metallo-beta-lactamases type, was first reported from *Pseudomonas aeruginosa* in 1990 and in today's date it comprises of more than 50 variants [1]. As a result, this resistance determinant has been reported in many enterobacterial as well as in non-enterobacterial isolates such as *Escherichia coli*, *Klebsiella pneumoniae*, *Enterobacter cloacae*, *Serratia marcescens*, *Pseudomonas*, *Acinetobacter* spp. and others. In consequence, the IMP type has become one of the most prevalent mechanism of carbapenem resistance in Japan and also in some parts of Europe and America and are found to cause severe community-acquired and health care-associated infections. So far, majority of the existing studies mainly focused on the antibiotic resistance around hospitals or among clinical isolates and as a result carbapenem resistant Enterobacteriaceae is already considered as endemic in many settings whereas in our study, we have reported the presence of *bla*<sub>IMP-4</sub> carbapenemases gene in sewage outfall located nearby a secondary hospital. The detection of this resistant determinant in environmental setting is really a growing concern and is gathering more attention as a potential source for human infection.

In this study, the *E. coli* carrying *bla*<sub>IMP-4</sub> was found to be resistant towards almost all the usually prescribed drugs and is genetically supported by the presence of multiple resistant determinants within the concerned isolate. The occurrence of a MDR phenotype in sewage water is may be due to the selective pressure in the natural environment which results into the acquisition of these acquired resistant determinants to evolve and to survive the challenging situation. The detection of *bla*<sub>IMP</sub> and *bla*<sub>CTX-M-15</sub> resistant determinants within a self-conjugable plasmid of *E. coli* directly indicated the significant concern of the spread of these resistance determinants among other susceptible phenotype leading it to a resistant one [6]. Also, the transfer of this resistant plasmid may lead a multidrug resistance (MDR) organism to an extremely drug resistant (XDR) phenotype and this results in to limited treatment option. Several previous studies revealed that IncN plasmid has a key role in *bla*<sub>IMP</sub> transmission and multispecies outbreak of *bla*<sub>IMP</sub> harbouring Gram-negative bacteria [6] including *P. aeruginosa* and other enterobacterial isolates but the present study revealed that the *E. coli* isolates carrying *bla*<sub>IMP</sub> was found to be conjugatively transferable through Inc F type plasmid. Inc F plasmid is reported as broad-host range plasmid and the acquisition of this resistant gene is may be due to the presence of

selective pressure of antibiotics in the environment [7]. This may be due to the increasing use of antibiotics in human medicine, agriculture and also in aquaculture settings which results into the release of large quantity of antibiotic molecules gathered in sewage, wastewater and marine environment. This release of antibiotics in the environment enhances the selective pressure which leads to the acquisition and evolution of resistant strain spreading their genes into many different species results into the detection of carbapenemases in sewage and waste water. Since some IMP-producing *Enterobacterial* strains, especially *E. coli* and *K. pneumoniae*, show low-level resistance or even sensitivity to carbapenems, it is quite difficult to detect in a clinical laboratory. So, in a report of Centers for Disease Control and Prevention, they have mentioned that IMP was identified in 0.4% of Gram-negative bacteria tested in 2017 and this under-detection of these strains may potentially contribute to the underestimation of the true burden of IMP carbapenemases. The current study underscores environment as a potential reservoir of carbapenem resistance, warrants a holistic approach to track and check dissemination of resistance in environment, human and agricultural settings.

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### Competing interests

The authors declare that they have no competing interests.

### CRediT authorship contribution statement

**Deepjyoti Paul:** Formal analysis, Writing – original draft. **Nurjhan Begum Mazumder:** Formal analysis. **Jayalaxmi Wangkheimayum:** Formal analysis. **Amitabha Bhattacharjee:** Supervision.

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